Héctor Somer

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SUMMARY

Backend developer with hands-on experience building RESTful APIs, ERP systems, and AI-agent tooling (MCP servers) using Java, PHP, and Python. Focused on clean architecture (hexagonal, layered) and automated testing, with leadership experience architecting a team project from the ground up.

EXPERIENCE

Software Developer

Realhost S.A. de C.V. **October 2025 - Present, Monterrey, Nuevo León, MX (Remote)**

- Designed and maintained 20+ client ERP systems using PHP, ensuring system stability and applying clean code patterns.
- Implemented straight-through processing (STP) using Laravel, Vite, and Tailwind CSS, minimizing manual data entry errors.
- Developed and maintained high-impact customer portals and self-service platforms, improving user engagement metrics.
- Deployed a complete application (frontend, backend, and database) on AWS EC2, working with Amazon RDS for relational database management.
- Built custom MCP (Model Context Protocol) servers in Python, connecting company systems to AI agent tooling for automated data queries and workflow actions.
- Built a chatbot integrating Chatwoot and Telegram channels, deployed via Docker with CI/CD pipelines in GitLab CI.

Back-End Developer

Life Agent S.A. de C.V. **October 2025 - April 2026, Ciudad de México, MX (Remote)**

- Collaborated with the team on 3 high-performance RESTful APIs using Java, Spring Boot, and Maven, implementing a layered architecture to separate routing from business logic.
- Managed robust database schema migrations and relational database version control using Flyway and MySQL, ensuring zero data loss during production deployments.
- Collaborated on integrating Firebase services to enable real-time data synchronization and secure user authentication.
- Used Google Cloud Storage to build secure, automated cloud file storage and management solutions.
- Wrote automated tests for each API using JUnit 5, Mockito, MockMvc, and AssertJ (via spring-boot-starter-test).

PROJECT

SIGEAN — Inclusive Secondary Education Management Platform

Ámbar Rojo Studios • October 2026 - Present

- Designed and built a REST API with Spring Boot 4.1, Java 21, Spring Data JPA, and Flyway, structured as hexagonal architecture split by feature (domain/application/infrastructure).
- Implemented JWT authentication with role-based access control for four user roles.
- Modeled a relational schema of 16+ tables with Flyway-managed migrations.
- Documented the API with Swagger/OpenAPI and used MapStruct for DTO-entity mapping.

LAMBI — Marketplace Platform (Team Project)

November 2024 - Present

- Led a 3-person student team as project manager and software architect, defining the backend architecture for a multi-vendor marketplace.
- Built the REST API in Python with FastAPI and SQLAlchemy, implementing JWT authentication and bcrypt password hashing.
- Designed routers covering buyer, seller, product, stand, and sales flows.
- Currently leading a full architectural rebuild of the platform.

AprendIA — Accessible Learning Mobile App (University-Sponsored Project, not coursework)

Universidad Politécnica de Chiapas • June 2026 - September 2026

- Collaborated on a cross-functional team of 5 students (2 backend, 3 frontend) and 2 faculty advisors, building an accessible literacy app for deaf and hearing users as an official university-sponsored initiative, not a coursework assignment.
- Built a microservices backend (API, core service, and gateway) in Spring Boot/Java 21, including a CRUD endpoint for role catalogs.
- Refactored entity models to use Lombok annotations and migrated DTOs to Java records for cleaner, more maintainable code.
- Containerized the system with Docker and Docker Compose (multi-stage builds), using PostgreSQL with the pgvector extension.
- Implemented JWT authentication, Spring Security, and health monitoring via Spring Actuator.
- Integrated Cloudinary for image asset management.

Cilantros — IoT Agricultural Monitoring System (Team Project)

Universidad Politécnica de Chiapas • March 2025 - April 2025

- Built Go-based microservices for user management and device communication, including registration, authentication, and role-based access.

• Integrated RabbitMQ (AMQP) for asynchronous messaging between services and WebSockets for real-time sensor data updates.
• Developed hardware integration services in Go and C++ to connect ESP32-based sensors for monitoring cilantro crop growth conditions.
• Implemented automated email notifications and deployed the backend services.

Secret Event — Anonymous Reporting DApp

Chiapas • October 2024 - October 2024

• Won 2nd Place at the Vara Side Hackathon among competing teams.
• Developed a Web3 DApp for secure, anonymous reporting of sensitive cases such as harassment and violence.
• Programmed smart contracts in Rust on the Vara Network blockchain to handle anonymous data submission securely.

EDUCATION

Bachelor's Degree in Software Engineering

Universidad Politécnica de Chiapas • Tuxtla Gutiérrez, Chiapas, Mexico • 2026 • 9.0/10 (≈3.7/4.0)

• Coursework completed (2023 - 2026) — currently completing mandatory professional internship (estadía); expected graduation: December 2026.
• AWS Academy Graduate - Cloud Foundations, Cloud Operations, Cloud Developing, Cloud Architecting | Amazon Web Services
• Cisco Networking & OS Basics - Networking Basics, Operating Systems Basics | Cisco.

SKILLS

Languages: Java, Rust, Python, Go, C++, PHP, Kotlin, Dart, TypeScript, JavaScript, SQL

Frameworks: Spring Boot, FastAPI, Gin, Laravel, Flask, Django, Express

Databases: MySQL, MariaDB, PostgreSQL, Flyway, Redis

Technologies / Tools: Docker, Docker Compose, RabbitMQ, Git, GitHub, AWS, Firebase, Google Cloud Storage, Vara Network, Hibernate/JPA, Ubuntu, SQLAlchemy, Gitlab, pytest, unittest, Maven, Bun, Node JS, Claude Code, Antigravity, Blockchain, Web3

Practices: ORM-based Persistence, Automated Testing, Hexagonal Architecture, RESTful API Design, Layered Architecture

Languages Spoken: Spanish and English (B1)